Model Solutions

Surface Mining

Ques-1 (a) Draw a balancing diagram for simple sidcasting operation of a dragline and derive the expression in terms of known parameter, for.

(i) height of spoil heap.

(ii) Reach of the dragline. (8)

(b) What is meant by MUF of a dragline and what is its physical significance? (2)

(c) What do you mean by "minimum tipping load" of a front-end-loader? Explain how the concept of minimum tipping load help in Selecting suitable bucket size in different job conditions (10)

Answer-1

The diagram for simple side cast is drawn on the next page →

(1)

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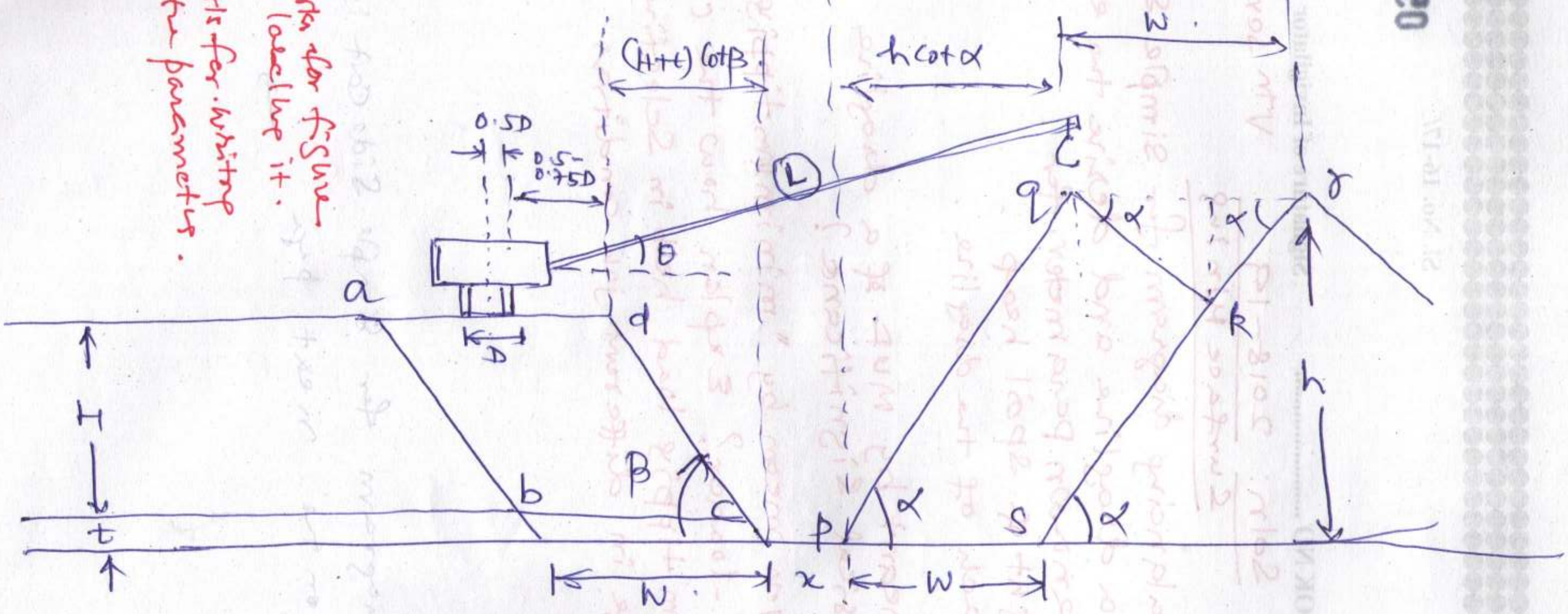
(2) marks for figure and labelling it.

(2) marks for writing all the parameters.



Side Casting Diagram (Balancing Diagram)

Bench Parameters

(1) H = Ht. of OB bench.
(2) W = Width of cut.
(3) t = thickness of Coal seam.
(4) β = Highwall slope angle.

Dragline Parameters

(5) D = Diametre of the tub (base) of D/L.
(6) L = length of boom.
(7) θ = Boom angle.

Dump parameters

(8) S = % swell of material.
(9) x = Clearance between highwall toe and spoil heap toe.
(10) α = angle of slope of spoil heap.
(11) h = height of spoil heap

The area abcd is in site area of OB.. This material is blasted and then it swells by a factor of $(1+\frac{s}{100})$. This swollen material is thrown in dump having shape of pqRS.

Since area of abcd = $H \times W$.

Area of dump (pqRS) $= H \times W\left(1+\frac{s}{100}\right)$ — ①

Now area of ||gm pqrs $= hW$ —— ②

area of $\Delta qRr = 2 \times$ Area(ΔqRu)

$$= 2 \times \left[\frac{1}{2} \times \frac{W}{2} \times \frac{W}{2} \tan\alpha\right]$$

$$= \frac{W^2}{4} \tan\alpha \quad \text{——} \quad ③$$

But ar (pqRS) = ar. (||gm pqrs) − ar(ΔqRr)

using Eqns ① ② & ③ we get

$$H \times W\left(1+\frac{s}{100}\right) = h \times W - \frac{W^2}{4}\tan\alpha$$

or. $$H\left(1+\frac{s}{100}\right) = h - \frac{W}{4}\tan\alpha$$ 2 marks for deriving Eqn ④

or $$\boxed{h = H\left(1+\frac{s}{100}\right) + \frac{W}{4}\tan\alpha.}$$ Eqn. ④

Now from the side cast diagram we have

$$\text{Reach} = 0.5D + (0.5 \text{ to } 0.75)D + (H+t)\cot\beta + x + h\cot\alpha$$

$$= (1 \text{ to } 1.25)D + (H+t)\cot\beta + x + h\cot\alpha$$

Substituting the value of h from Eqn (4) we have.

$$\text{Reach} = (1 \text{ to } 1.25)D + (H+t)\cot\beta + x + \left[H\left(1+\frac{S}{100}\right) + \frac{W}{4}\tan\alpha\right]\cot\alpha$$

or. $$\text{Reach} = (1 \text{ to } 1.25)D + (H+t)\cot\beta + x + H\left(1+\frac{S}{100}\right)\cot\alpha + \frac{W}{4} \quad \text{——— (5)}$$ $1\frac{1}{2}$ marks for Eqn 5

again we have

$$L\cos\theta = R - 0.75D$$ $\frac{1}{2}$ marks for showing $L \simeq R$.

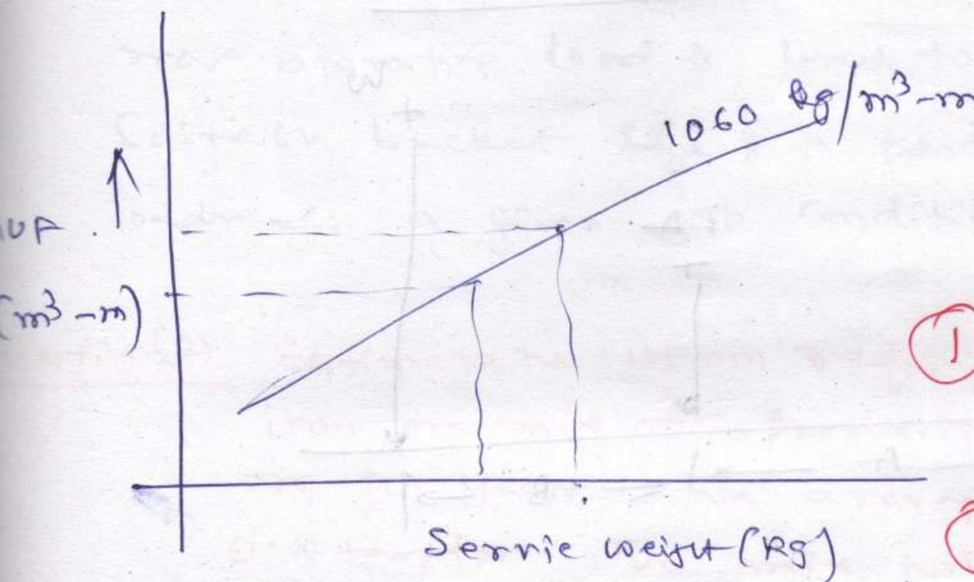
Considering θ to be 35^0 to 39^0 (most common)

we get $L \simeq R$.

Eqn (5) is the reach of the dragline &

Eqn (4) is the height of the spoil heap

b) MUF — Maximum usefulness factor



(1) marks fr. graph & labelling ~~explanation~~

(1) marks fr. explaing MUF & its defn.

it is the measure of the stability of a m/c. & is given by

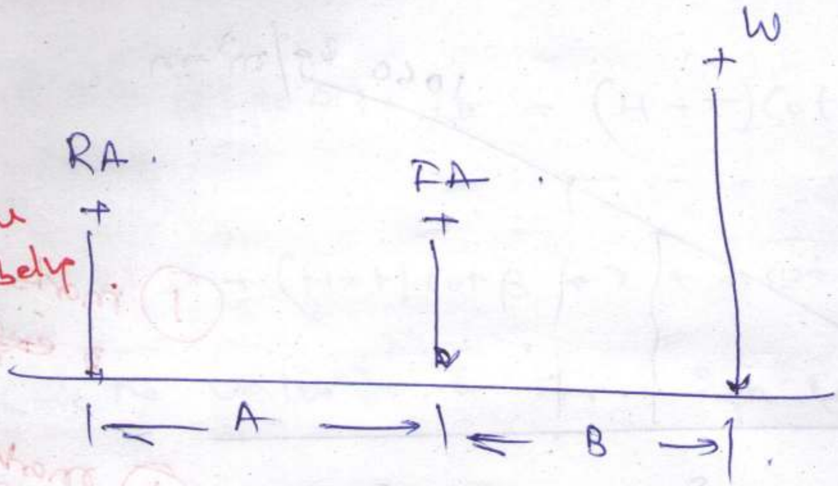
$$MUF = B_c \times R \approx B_c \times L$$

After the required bucket capacity and reach are estimated for the given application, minimum MUF required is calculated. This is followed by calculation of the min reqd service wt of the m/c. (wt of m/c + wt of ballast) using the MUF relationship curve.

Thereafter few models from different manufacturer's reference having service weight more than the minimum reqd. service wt. is selected, from which the optimum one is chosen based on the economic consideration.

(C). Concept of tipping load

2½ marks for figure & its labelling



(1) marks for MTL explaining its defn

Minimum tipping load is the measure of the Stability of front end loaders at minimum tipping load.

$$A \times RA = B \times W$$

Where:

RA = m/c wt acting on the rear axle.

W = wt of bucket and load

A = distance bet two axle of the m/c.

B = dist bet front axle of m/c & the Centre line of the bucket in extended Condition.

(2½) marks for explanation & labels.

Generally, the min tipping load is about 70-80% of the m/c wt. depending on its Config.

Max. operating load < 50% of min tipping load.

(2½) marks for indicating the two points.

(2½) marks for explaining the understanding of application.

The concept of minimum tipping load & max operative load is used to find out the suitable bucket size of a particular front end loader for a given job condition.

Question (2) Estimate the HEMM required in a surface iron-ore mine for producing 10 million ton of ore per year. The average stripping ratio is 0.8 ton/ton. The mine will be worked by a shovel dumper combination with a bench height of 12 m. (Assume ~~of~~ relevant data wherever necessary and mention them)

Solution-(2)

Given Condition

(i) Production of ore per year = 10×10^6 ton

(ii) Average stripping ratio = 0.8 ton/ton

(iii) Bench height = 12 m (½) marks for. write the given condition.

Assumption.

Mining parameters (½) marks for realistic assumptions.

(i) Bank density of ore = 3.6 ton/m3

(ii) Bank density of waste = 2.8 ton/m3

Shovel parameters

(i) Bucket capacity = $5 m^3$

(ii) Bucket fill factor = 0.8

(iii) Swell factor = 25 %.

2 marks for realistic assumption

(iv) Standard cycle time of shovel = 30 s. (90° swing)

(v) Overall efficiency = 0.56

(vi) Working Hrs = 300 × 3 × 8 = 7200 Hrs

Dumper - parameter :

(i) rated capacity of dumper = 50 ton.

(ii) Spotting time of dumper = 0.3 min.

2 marks for realistic assumption.

(iii) Dumping time of dumper = 1.5 min

(iv) Load travel speed of dumper = 20 km/hr.

(v) Empty travel speed of dumper = 30 km/hr.

(vi) Haul distance (one way) both ore + waste = 1 km.

Calculation

Production of ore per year = 10×10^6 ton.

Handling of ~~w~~ waste for yr = $0.8 \times 10 \times 10^6$

$= 8 \times 10^6$ ton.

Total material to be handled per year } = 18×10^6 ton.

average density of composite (ore & waste) } $= \dfrac{(1+SR) \times \rho_o \times \rho_w}{(\rho_w + SR \cdot \rho_o)}$ ton/m3.

3 marks for evaluating the average density.

$= \dfrac{(1+0.8)(3.6)(2.8)}{(2.8 + 0.3 \times 3.6)}$

$= 3.279$ t/m3.

Estimating the No of shovels

Required material handling per year (te) = 18×10^6

Average density of ore & composite = 3.279

4 marks for calculating No. of shovel.

Effective production per shovel per year (te) } $= B_c \times B_f \times S_f \times \dfrac{3600}{t_c \times F_s} \times 7200 \times \eta \times \rho_{av}$

$= 5 \times 0.8 \times \left(\dfrac{100}{100+25}\right) \times \dfrac{3600}{30 \times 1} \times 7200 \times 0.56 \times 3.279$

$= 5076836.35$ te.

Calculation

No of shovels Require $= \dfrac{18 \times 10^6}{5076836.35}$

$= 3.546$

$$\boxed{\text{Actual no of Shovels Reqd} = 4}$$

Calculating No of Dumpers.

Effective Cap. of dumper (tc) = 50

ton per pass of Shovel $= B_c \times B_f \times S_f \times Q_{av.}$

$= 50 \times 0.8 \times \left(\dfrac{100}{100+25}\right) \times 3.279$

$= 10.493$

No of passes Reqd to fill the dumper $\Big\} = \dfrac{50}{10.493}$

$= 4.77$

(2) marks for calculating No. of passes.

Actual No of pass reqd to fill the dumper $\Big\} = 4.0$

Total loading time of dumper (min) = 2

Spotting time (min) = 0.3

Dumping time (min) = 1.5

Load travel time $= \dfrac{60 \times 1}{20} = 3.0$

Empty travel time $= \dfrac{60 \times 1}{30} = 2.0$

Total cycle time of dumper (min) = 8.8.

No of Dumpers reqd = $\frac{\text{Total time}}{(\text{Spotting time + loading time})}$

4 marks for calculating No of dumps. = $\frac{8.8}{(2+0.3)}$ = 3.83

~~2 Summarising~~.

Actual No of dumps per shovel = 4

So Total No of dumpers Reqd = 4x4 = 16

Summary of HEMM

Item	Capacity	Qty	Total	Remark	Spare
Shovel.	5 m3.	4.	6		2
Dumper.	50 te	16	24		8
Drill.	250mm	4	6		2
Dozer.	250/350HP.	8	12		4
FEL	3.2 m3.	2	3		1
Motor grader.		1	2		1
Crane.		1	2		1
Water sprinkler.	5Kl.	1	2		1

2 marks for summarising the table.

if unrealistic wrong assumption is used in, but calculation steps are OK. each step will be evaluated at 50%. marks.

Q. 3. (a) What is a "box-cut"? What are its objectives? With the aid of suitable sketches, describe the different types of box-cut and also indicate their respective applicability and limitations.

(b) Discuss briefly the factors that influence the choice of location of box-cut in case of sub-surface deposits.

Ans:- Box-cut is the initial/first cut given for the physical development of a mine. The name "box-cut" has been attributed to this cut as this cut generally looks like a top open box having an inclined floor & walls on three sides (front wall & two side walls) in case of opening up of sub-surface deposits.

2 marks for defn & expln

Objectives of a box cut

- to reach the ore body/coal seam
- to provide smooth entry to the pit.
- to provide space for development of workings (production benches.

2 marks for enumerate the objective.

Types and applicability of Box-cut

(1) Internal box cut:

When the box cut is located fully or partially on mineralised zone, it is called an internal box-cut.

(2)

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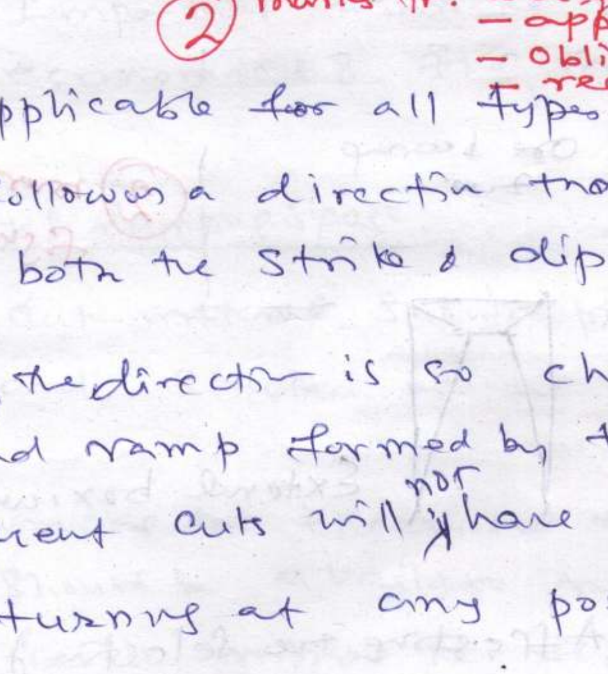
(2) marks fr. - defn
- applicibility.
- obliquity reasons :

This is applicable ~~for~~ all types of deposit.

The cut follows a direction that is usually oblique to both the strike & dip direction.

Generally, the direction is so chosen that the haul road ramp formed by this cut and subsequent cuts will not have unnecessary steep turning at any position.



(2) marks for figure.

Internal Box cut.

External Box-cut

When the box cut is placed totally outside the mineralized zone, it is called an external box-cut.

(2) marks for defn expln & applicability.

This is applicable only for shallow and gently dipping bedded deposits.

This cut is generally located at the middle of the rise-most side.

2 marks fr. figure.

Extenal boxcut.

Factors Affecting the selection of Box-cut.

Selection of location of box-cut for any surface mine is of utmost importance as it has far reaching effects on the economy of the mine.

The location should be so chosen that there is a min of foreseeable trouble with uncovering of the ore body and a max. of economy.

Factors

(1) Site accessibility: easily assessible/reachable for men & machinery.

(2) Minimum Excavation Requirement

- material to be excavated to reach the ore body will be as less as possible.
- this ensures early production of ore from the mine.
- Important from the view point of the economics of the mine operation.

(3) Availability of dumping space:

- The cut material should preferably be handled only once and stacked off the mineral bearing zone.
- for dumping the excavated material, sufficient space should be available nearby.
- this will improve the economy of transport of cut material.

8 marks for each point.

(4) Pit haulage plan:

- if the box cut is planned to be a part of the pit haulage system, it must match with the haulage plan for the complete pit.
- the roads as far as practicable, should be put in but once and used as much as possible.

(5) Overall mine plan:

- The box cut location should be so chosen that it matches to/in the overall mine plan including the waste dump locations.
  - Sites for infrastructural facilities.
  - Sites for other facilities.
- it should not hinder the pit progress/expansion at a later date.
- the location should be such that the overall economy of ore and waste transport over the whole mine life is maintained.

(6) Water Condition.

- Surface drainage must be diverted, preferably by gravity.
- if possible, the location of the cut should be so chosen that the problems of accumulation of water from surface water and/or ground water are not there or minimum.

(7) Geological Disturbance

- free of any geological discontinuity and structural disturbances.

(8) Reclamation Requirement:

- The reclamation requirements may dictate that the box-cut material should be dumped / emplaced at particular location.

Ques-(4) (a) With the help of suitable sketches, describe the lateral block cut method operation of bucket wheel excavator.

(b) Calculate the theoretical hourly output for a bucket wheel excavator from the following given data:

(i) No of buckets on wheel = 10.

(ii) Individual bucket cap = 1200 lts.

(iii) Wheel diameter = 9 m

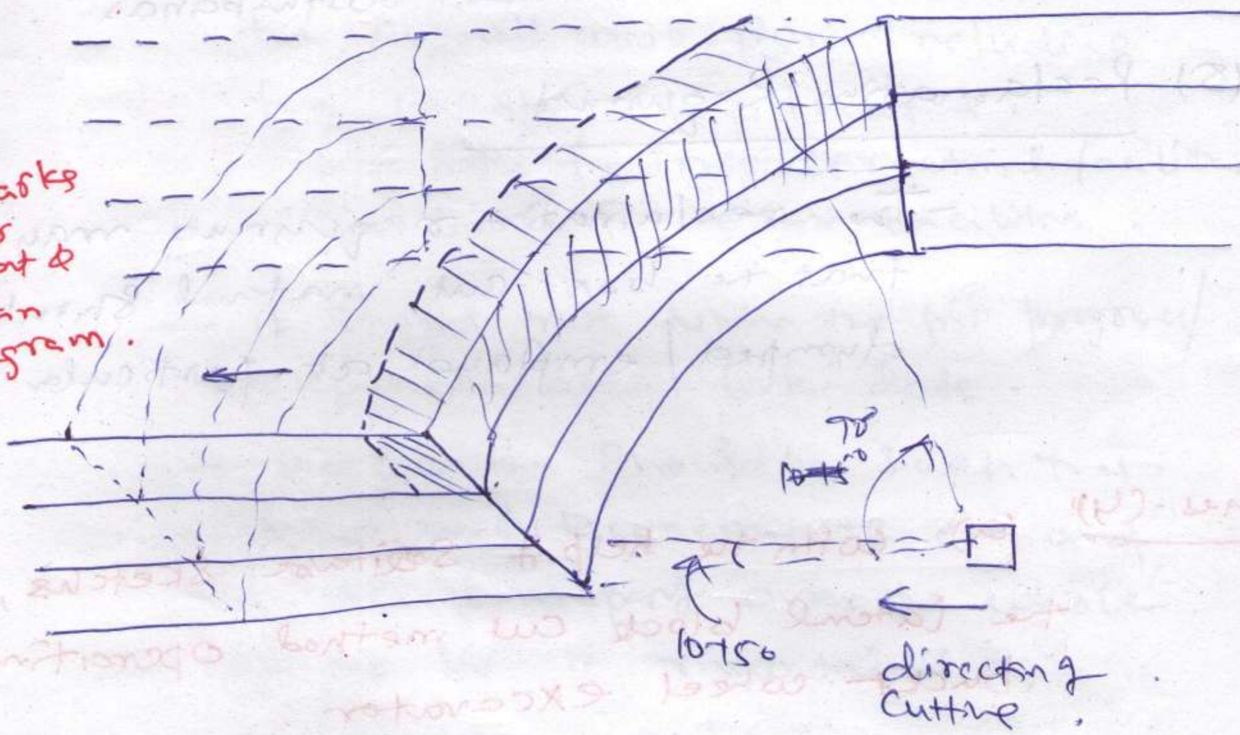
(iv) Cutting speed at the knife edge = 2.1 m/s.

(v) % swelling of the material to be excavated = 25

Ans- 4 (a)

Lateral block / Half Block Method

3 marks for neat & Clean diagram.



- The machine travels parallel to the highwall with crawlers offset.
- The machine cuts long deep benches or terraces along the highwall keeping both the highwalls (old & new) on the same side.

5 marks for relevant explanation

and hence the method is also known as ½ block method.

- height of the terrace = 0.5 to 0.7 D
  where D = Dia of bucket.

- in this method of excavation the angle of swing of the cutting boom into the highwall is for $10°$ to $15°$ through $90°$ to the direction of the travel.

- for this operation the BWE is generally required to be equipped with long cutting boom.

- The m/c cuts the whole bench height in required no. of slices starting at the top.

There is significantly more travel time required than that with the full block method, reducing the actual digging time and increasing the crawler wear. However, the method is mostly used in case of selective mining of interbedded ore and waste.

(b). The theoretical productivity of BWE Q_{th} is given as follows.

$$Q_{th} = I \times N_{BD} \times S_f \times 60 \ m^3/hr.$$

I = individual bucket cap (Normally Lts, convert to m^3)

(1) marks for self explanatory formula with description

N_{BD} = No of buckets discharge per unit time.

S_f = Swell factor $= \left(\frac{100}{100+S}\right)$

$$N_{BD} = \frac{60 \times V_c}{\pi D} \times B_N .$$

1 marks for Showing calculation of N_{BD} with expln

V_C = wheel speed. (tangential speed at cutting edge)

B_N = No of buckets attached to wheel (6 to 20).

D = dia of cutting wheel. (m)

$\therefore$ No of buckets discharged per unit time (min)

$$N_{BD} = \frac{60 \times 2.1}{\pi \times 9} \times 10$$

2 marks for calculating in m^3/hr

one marks deducted for not giving Q_{th} in (m^3/hr)

$$= 44.56338$$

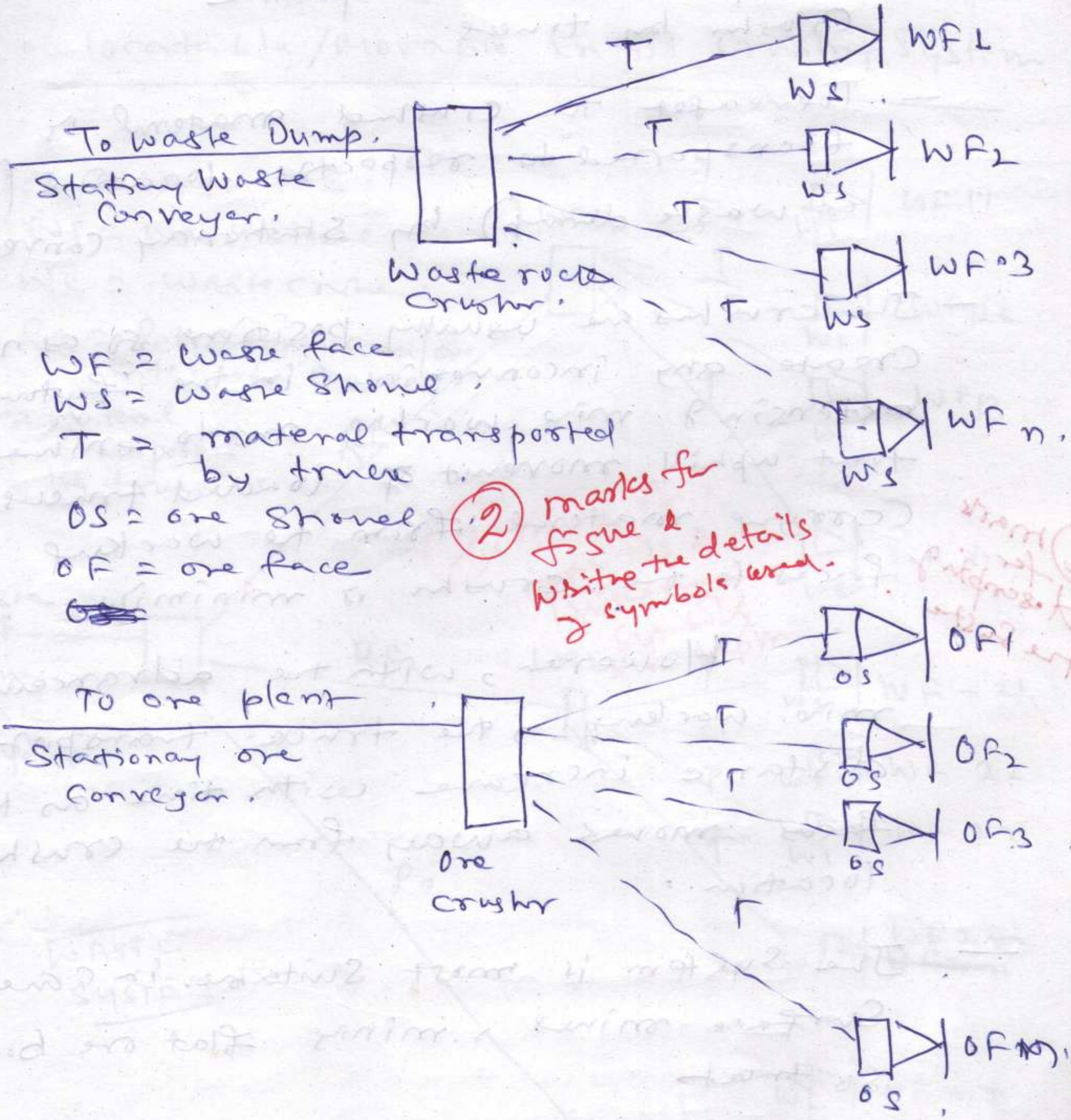
$$\therefore Q_{th} = \left(\frac{1200}{1000}\right) \times \left(\frac{60 \times 2.1}{\pi \times 9} \times 10\right) \times \left(\frac{100}{100+25}\right) \times 60 \ m^3/hr$$

$$= 2566.85 \ m^3/hr$$

(Ques-5) With the help of schematic diagram describe the different types of in-pit crushing and conveying system that may be adopted in surface mine.

Ans-(5) Different types of In-pit Crushing Systems.

(I) Stationary/Permanent In-pit Crushing System.



WF = Waste face
WS = Waste Shovel,
T = material transported by truck
OS = ore Shovel
OF = ore face
~~OS~~

(2) marks for figure & write the details of symbols used.

→ In this type of system, large capacity Crushers (separate for ore and for waste) with proper foundation are installed in permanent locations at or near the pit floor level.

- The material excavated and/or loaded by Shovel ; from all the working faces (of same type ie ore or waste) are transported to the respective Crusher by trucks

- Thereafter the crushed material is transported to respective location (ore or waste dump) by stationary conveyors.

- The crushers are usually positioned so as not to create any inconvenience in the future extension of mine working and to ensure that uphill movement of loaded trucks carrying material from the working faces to the crusher is minimized.

② mark for describing the system

However, with the advancement of mine working, the truck transport distance increase with time as the faces moves away from the crusher location.

- The System is most suitable in case of surface mines, mining flat ore bodies so that

(i). In course of operations, the pit floor depth remains essentially same

(ii) Large quantities of ore can be transport over a long distance that may or may not be associated with

Considerable vertical lift.

- The benches on one side of the pit (or at least a part of it) have reached their ultimate position and became stationary.

(2) Relocatable/Movable In-pit Crushing System.

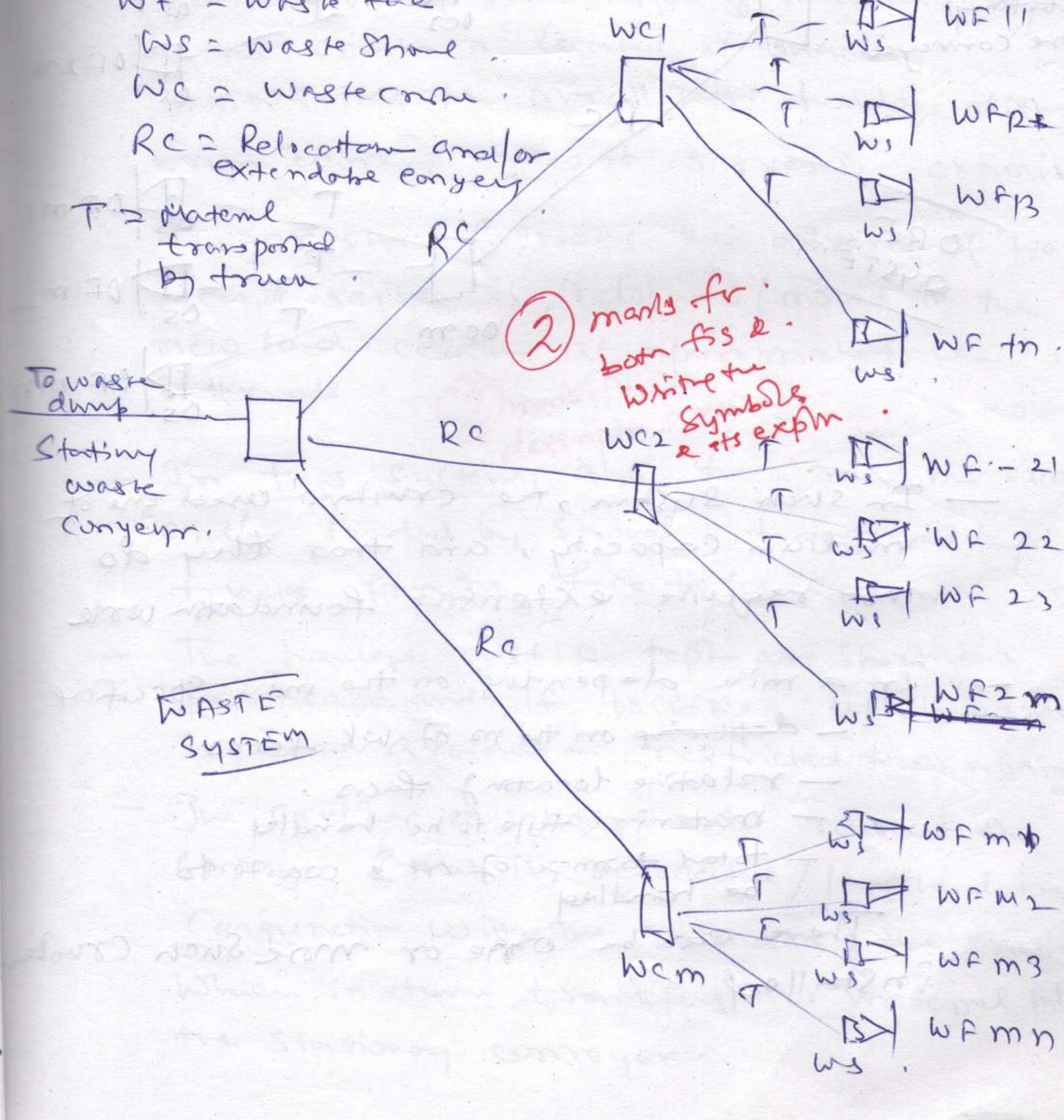
WF = Waste face

WS = Waste shovel.

WC = Waste crusher.

RC = Relocation and/or extendable conveyor

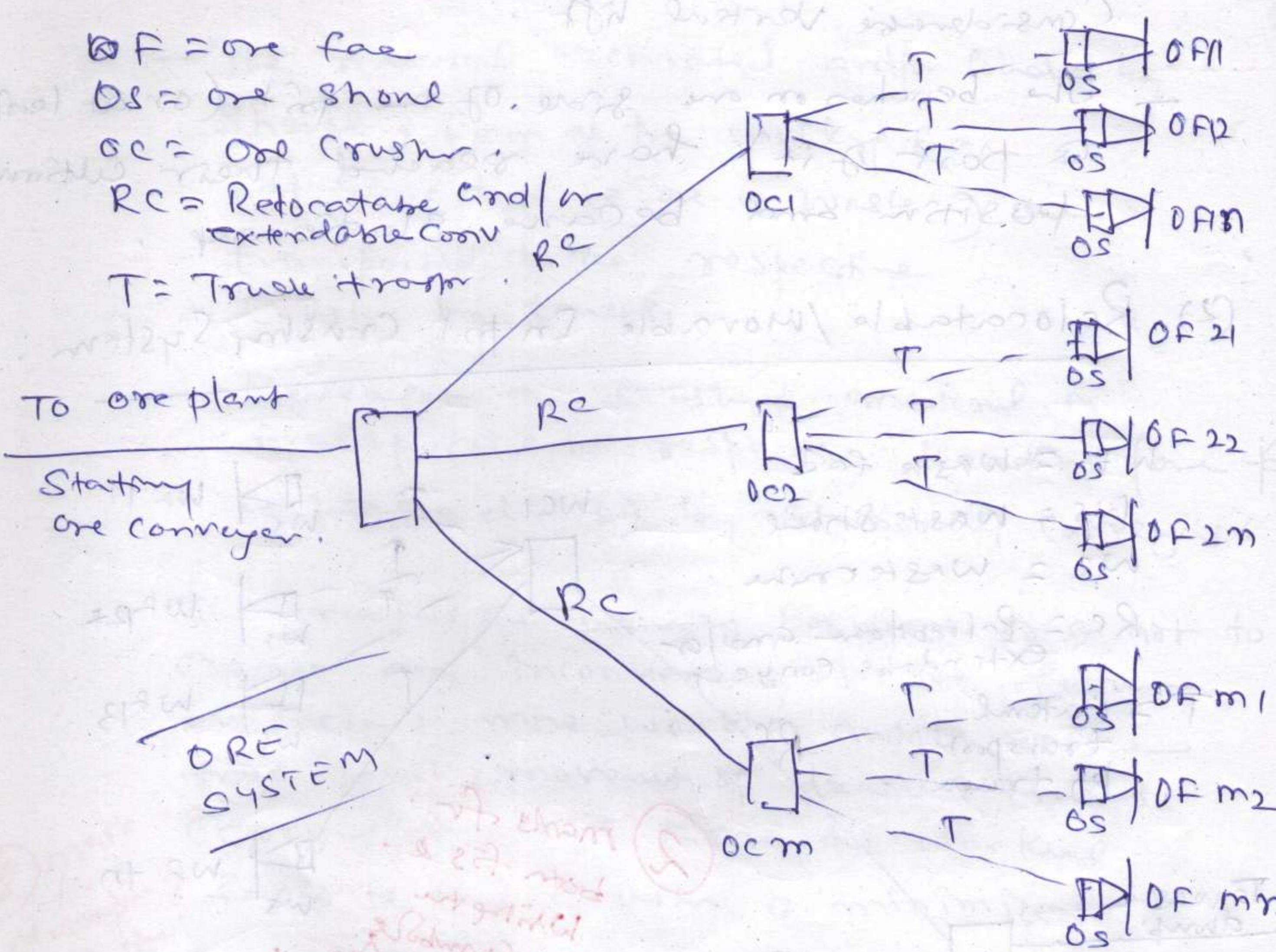
T = Material transported by truck.



(2) marks for both figs & write the symbols & its expln.

WASTE SYSTEM

OF = ore face
OS = ore shovel
OC = ore crusher
RC = Relocatable and/or extendable conv
T = Truck transfer



— In such system, the crushers used are of medium capacity, and that they do not require extensive foundation work

— In a mine depending on the mine specifics
- ~~depending~~ on the no of wk. faces.
- relative location of faces.
- material type to be handled
- total tonnage of ore & waste to be handled

There can be one or more such crusher installed

- Generally, each Crusher Serves a group of working faces excavating same type of material (ore/waste) and the crusher is located at such a position that the overall ~~dist~~ haulage dist from working faces it serves is optimised.

- The crusher following the advance of working faces it serves is relocated/moved to the new load centre at appropriate time intervals.

2 marks for: description & details.

- In this system, also, the material excavated and/or loaded by Shovel is transported by trucks from the face to the crusher.

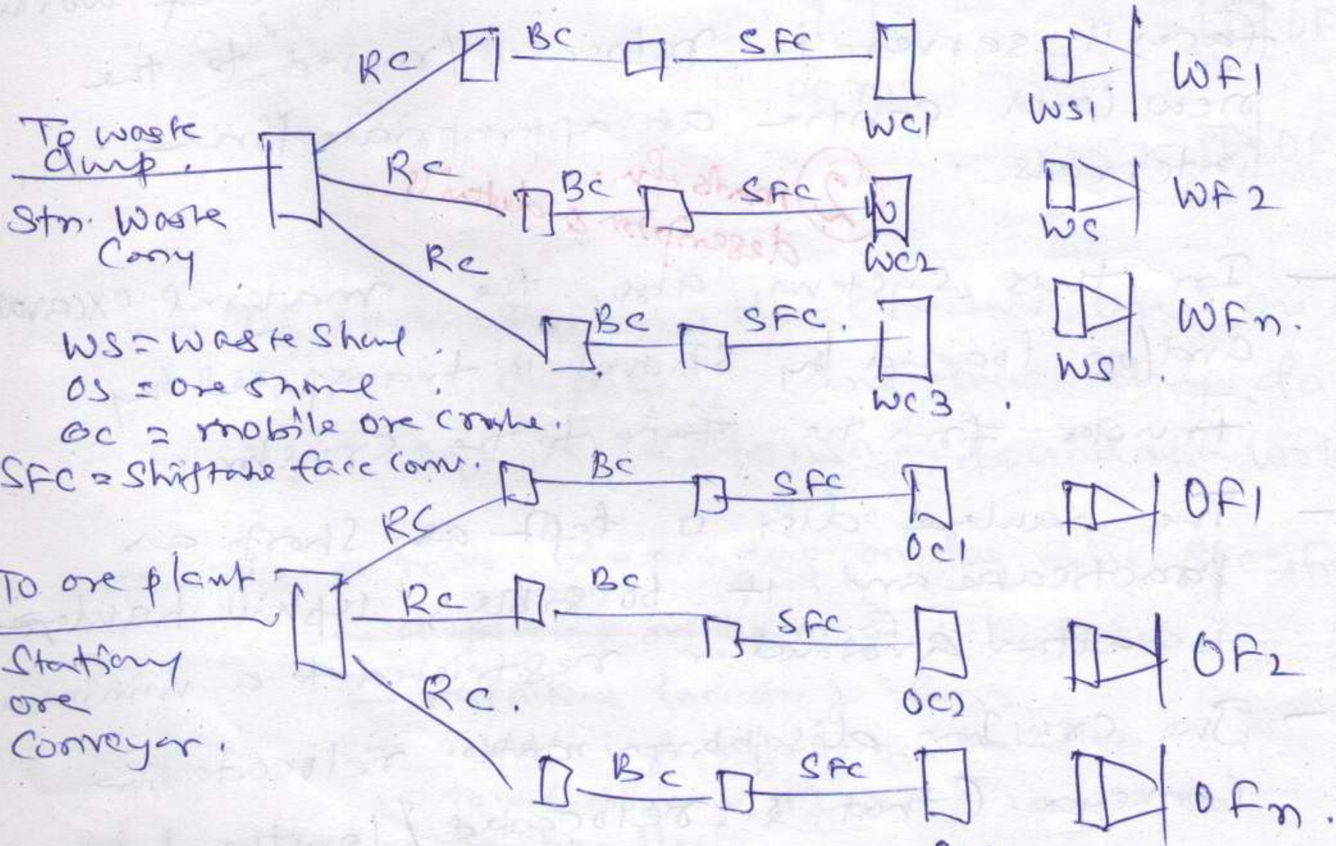
- The haulage dist is kept as short as practicable and, if possible, uphill haulage is either avoided or restricted to a minimum.

- The crusher discharges onto relocatable conveyor (that is relocated/lengthened in conjunction with the relocation of the crusher) which in turn transfers the material into the stationary conveyor.

- The System is suitable for a mine where -

- The production is medium to large.
- → The working faces are distributed in number of benches.
- A stationary in-pit crushing system is not suitable.

Mobile In-pit Crushing System



Bc = Bench conveyor

RC = Relocatable and/or extendable Conv,

2 marks for figure.

- In such a system, the crusher is fully mobile and is located at very near to the face.
- The crusher is fed directly by an excavator. (rope shovel / hydraulic shovel / FEL)
- The crusher discharges either directly or via a beltwagon (or conveyor bridge in some cases) onto the shiftable face conveyor.
- Finally the material is conveyed to the respective destination (ore plant / waste dump) through a network of belt conveyors.
- The crushers are generally of small capacity sufficient to handle the production from the face where it is located.

This system is applicable for mine where only a few fast moving working faces exist and the excavators stay permanently at the same face.

2 marks for explanation.

PIPARWAR - OCM.

(6) Estimate the no. of drills required to achieve a production of 5 million ton/year of coal from a surface mine having coal seam of thickness 10m. The coal seam is flat and horizontal and the stripping ratio of the mine is 3 m3/ton. (Assume relevant data wherever and mention them).

Ans:-

(2) marks for calculating the OB thickness

$$\boxed{\text{OB thickness} = SR \times \left(\text{Thickness of Coal Seam}\right) \times \rho_{coal}}$$

$$\text{OB thn} = 3 \times 10 \times 1.6 = 48\text{ m.}$$

∴ 4 Benches of 12 m each.

Given Condition:

(1) marks for mentioning the given conditions & assumptions of realistic density

1. Coal production per year = 5×10^6 te.
2. SR = 3 m3/ton
3. Coal seam thickness = 10 m.
4. Insitu density of Coal = 1.6 ton/m3.

Assumptions.

	Coal Seam	OB Bore
1. Drill dia	150	250
2. Burden	4.5	6
3. Spacing	6.5	8
4. Bench height	10	12 m
5. Length of drill rod	6	7

	Coal	OB
6. Time to set drill m/c (min)	6	7
7. Time to set drill rod (min)	3	3
8. Rate of drilling (min/m)	2	4
9. Subgrade drilling	0	10 %
10. overall utilization	0.6	0.6
11. Drilling hrs per (yr)	3000	3000

2½ + 2½
for realistic assumptions for OB & coal

Calculation:-

Estimation of Drilling requirement/year.

Volume of coal handling/year = 3125000

Vol of coal handled/hole (m^3) = 292.5

No of holes Reqd/year = 10683.76

Actual No of holes Reqd/yr = 10684

No of feed Reqd = 1.66 ≈ 2

time to drill one hole = 29 min.

3 marks for calculating time to drill

Capacity of drill m/c in coal.

Actual holes / hr = 2.0690.

Actual Hole per yr = 3724.14.

No of drills = 2.86 (3) marks for calculating no of drills.

Actual No of drills = 3.

Estimating OB. Drills

a) Drilling requirement per year

Vol of OB handling / yr (Prodn x SR) = 1,50,00,000.

Vol of OB handled / hole = 576

No of holes Reqd = 26041.667.

Actual No of holes Reqd = 26042.

Depth of drilling in OB = 13.2 m.

No of feed Reqd = 1.88 ≃ 2.

No of feed change Reqd = 1

(3) marks for calculating time to drill

time to drill one hole = 62.8 mne

Cap of drill mle in og

if steps are correct & using data from upstream is used it will be evaluated at 50%.

Actual holes/hr = 0.955

Actual hours/yr = 1719.75

No of drill reqd = 15.14

Actual no of drills = 16

(3) marks for calculating the No of drills including summary.

∴ OB - Drills = 16
Coal Drills = 3